CIS 641: In-Class Activity (System Request)
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System Request: Online Ordering and Kitchen Inventory System

Project Sponsor: Maria Lopez, Owner and General Manager, Corner Table Kitchen (a family owned restaurant near campus in Allendale, MI)

Business Need: Corner Table Kitchen takes orders by phone, at the counter, and through two third party delivery apps. Phone orders are written on paper tickets, and the delivery app orders come in on separate tablets, so staff have to retype them for the kitchen. This leads to wrong or missed orders during busy hours. The delivery apps also keep 25 to 30 percent of each order as commission. Inventory is counted by hand once a week, so the kitchen sometimes runs out of popular items or throws away food that was over ordered. This project has been started to give the restaurant its own online ordering channel and to connect all orders and stock levels in one place.

Business Requirements:

- Customers can view the menu and place pickup or delivery orders from the restaurant's own website or phone
- Customers can pay online or choose to pay at pickup
- All orders (online, phone, counter, and delivery app) show up on one kitchen display screen in the order they were received
- Staff can mark items as sold out, and the online menu updates right away
- Stock levels go down automatically as orders are completed, and the manager gets an alert when an item is running low
- The owner can see daily and weekly reports on sales, best selling items, and busy hours
- Returning customers can save their details and earn simple loyalty points

Business Value:

We expect this system to cut down on order mistakes, shorten wait times at the counter, and give customers an easier way to order directly from us instead of through the apps. Having real inventory numbers should also help the owner order supplies based on actual sales instead of guesses. Customer satisfaction and repeat business should go up, although these are hard to put a number on at this stage.

Conservative estimates of tangible value per year:

- $7,200 saved on delivery app commissions (about $6,000 a month currently goes through the apps at a 25 percent fee; if 40 percent of those orders move to our own site, fees drop by about $600 a month)
- $4,700 saved by cutting remade orders in half (right now around 15 orders a week are remade, at about $12 in food and labor each)
- $1,200 saved on food waste (about $400 a month is thrown out today; we expect a 25 percent reduction)

- $18,000 in new sales from customers who order online during hours when the phone line is too busy to answer (estimated at $1,500 a month)

Special Issues or Constraints:

- The owner's budget for the first version is about $15,000, so the project should use existing tools where possible, such as a hosted payment service, rather than building everything from scratch
- Card payments must be handled by a PCI compliant payment provider. The restaurant should never store card numbers on its own system
- The online ordering part should be working before the spring semester starts in January, since student orders make up a large share of business during the school year. Inventory tracking and reports can follow in a second phase
- Most kitchen staff are part time students, so the kitchen screen has to be simple enough to learn in one short training session
- The restaurant will keep using the delivery apps for now, so the system needs a way to bring those orders into the same kitchen queue